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Dual CNN and ViT experts fusion for open set recognition

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ABSTRACT

Open set recognition (OSR) aims to enable deep neural networks to accurately classify known categories while identifying instances belonging to unknown categories. Existing OSR methods often adopt discriminative approaches, with convolutional neural networks (CNNs) serving as the core architecture. However, vision transformers (ViTs), transformer-based architectures developed for computer vision, have demonstrated superior capabilities in capturing global contextual information compared to CNNs. This study proposes a multi-expert fusion network that integrates ViT and CNN architectures for OSR, termed the CNN and ViT Experts Fusion (CVEF) model. CVEF integrates multiple ViT and CNN experts, enabling the capture of both global and local image features while adaptively fusing their outputs to distinguish between known and unknown categories. Analysis reveals that ViT and CNN experts attend to distinct aspects of images. Extensive experiments on standard OSR benchmarks demonstrate the effectiveness and robustness of the model. The proposed approach holds potential for broader applications in image processing and recognition tasks. The code for this work is available at https://github.com/xulaupuz/CVEF.

1. Introduction

Over the years, deep neural networks (DNNs) have achieved, and in some cases surpassed, human-level performance in various tasks, including image recognition He et al. (2015), image segmentation, and image classification. Conventional image classification has long served as a fundamental approach for recognition tasks Zhao et al. (2024). This approach operates under the assumption that the categories encountered during training and testing are identical and fixed, thus implying that the model is only required to classify data into a limited set of known categories observed during training. However, in real-world applications, this assumption often proves unrealistic, particularly in dynamic and open environments where models often encounter numerous previously unattested categories Yoshihashi et al. (2019). This limitation of conventional classification results in poor performance in open set recognition (OSR) scenarios, as the model is unable to identify unknown categories, often leading to the misclassification of unknown instances as known classes. Thus, conventional classification presents challenges in applications where the dynamic handling of new or unknown categories is essential, thereby highlighting the need for robust OSR methods, particularly in high-stakes decision-making processes such as industrial quality control and public safety surveillance.

Models designed for OSR perform two key tasks: correctly classifying samples originating from recognised categories and accurately detecting those belonging to unfamiliar or previously unattested classes Scheirer et al. (2013). Recent work on OSR has predominantly focused on two approaches: generative methods and discriminative methods. Generative methods aim to model the distribution of known classes, allowing the system to detect instances that do not conform to these distributions as unknowns Geng et al. (2021). Methods such as generative adversarial networks (GANs) and variational autoencoders (VAEs) are often employed, as they are capable of capturing complex data distributions and identifying anomalies or outliers that may belong to unknown classes. However, owing to their high computational cost, particularly when handling high-dimensional data, generative models often present scalability challenges. As the number of known classes increases, maintaining accurate representations for each class becomes increasingly difficult Vaze et al. (2021). Furthermore, these methods may not always generalise effectively to entirely new or diverse unknown data, potentially resulting in misclassification Xu et al. (2023).

Unlike generative methods, discriminative methods focus on optimising the decision boundaries between known classes, often incorporating mechanisms to recognise and reject instances that fall outside these boundaries. Discriminative approaches typically employ DNNs

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with additional components or strategies, such as thresholding mechanisms or OpenMax layers, to identify unfamiliar data. Although discriminative methods are generally more efficient and easier to scale compared with generative methods, they may struggle to handle extreme outliers or rare unknowns effectively Geng et al. (2021).

More recently, the transformer architecture applied to computer vision, the vision transformer (ViT) Dosovitskiy et al. (2020), which leverages the self-attention mechanism originally developed for natural language processing, has attracted substantial interest for its scalability and robust performance. Despite their robust performance in computer vision, existing OSR methods predominantly utilise convolutional neural networks (CNNs) as the core architecture, including ResNet-based autoencoders and multi-expert systems. This approach raises the question of whether ViTs can be effectively applied to OSR tasks.

In this work, a novel approach to OSR is introduced by utilising ViTs for the task. A multi-expert system combining multiple ViT models with CNN experts is proposed to leverage the strengths of both architectures: ViTs offer comprehensive global contextual understanding, while CNNs provide localised feature extraction, which is particularly effective for fine-grained details. Through the integration of outputs from both ViT and CNN experts in a model termed CNN and ViT Experts Fusion (CVEF), this approach facilitates a more comprehensive decision-making process, thereby enhancing the model's ability to accurately and robustly differentiate between familiar and novel categories.

Extensive experimentation on numerous standard and large-scale OSR benchmarks demonstrates that the proposed technique outperforms existing discriminative and generative models, achieving improved area under the receiver operation curve(AUROC) scores and excellent overall performance. CVEF is simple to implement and scalable with respect to both the number of experts and the complexity of the ViT and CNN components. In summary, the contributions of this study are as follows:

1. An efficient and effective method for OSR, termed CVEF, is proposed. This method integrates CNNs and ViTs within a multi-expert network architecture, combining their outputs for image OSR. CVEF leverages the strengths of both architectures by capturing both global contextual information and localised features, thereby enhancing OSR performance.
2. This work presents a universal framework capable of incorporating various neural network architectures as experts, making it adaptable to different datasets and tasks. The modular design facilitates the seamless integration of new expert models, supporting both scalability and diversity.
3. Visualisation of expert attention mechanisms reveals that ViT experts capture global information but may ignore critical local features. Conversely, CNN experts focus on local features, potentially neglecting global context. Thus, relying solely on either expert type can lead to suboptimal performance.
4. According to comprehensive experimentation on both standard and large-scale benchmarks, the proposed method consistently outperforms discriminative and generative baseline approaches.

2. Related work

The OSR challenge, conceptualised by Scheirer et al. (2013) in terms of open space risk, has been substantially improved through the leveraging of the robust representation capabilities of DNNs. Current improvements can be broadly classified into generative-based approaches and discriminative-based approaches, both of which contribute to enhanced OSR performance.

2.1. Generative methods

Generative networks are deployed to create unknown samples to help the discriminator learn more effectively. G-openmax Ge et al. (2017) and OSRCI Neal et al. (2018) utilise GANs Goodfellow et al. (2014) to generate unknown class samples, while OpenGAN Kong and Ramanan (2021) generates synthetic features directly rather than producing fake images, achieving superior performance with lower computational cost. Autoencoder (AE)-based methods Kingma and Welling (2014) typically reconstruct known classes and generate corresponding representations of these classes. Using a conditional variational autoencoder (CVAE), the two-stage method C2AE Oza and Patel (2019) obtains class-conditional representations while training an expert block to proficiently detect unattested entities, thereby minimising reconstruction errors. CapsNet Guo et al. (2021) integrated a CVAE with a capsule network to align outputs with a predefined distribution. RPL Chen et al. (2020) differentiates unknown samples from known counterparts based on deviations from reciprocal points. Building on RPL, ARPL Chen et al. (2021) enhances the model's discriminative capability by generating deceptive samples using GANs.

Generative methods aim to learn different distributions for known classes or to generate synthetic unknown samples that are similar to the known classes. Although recent advancements have demonstrated remarkable performance, these models may struggle to discriminate reliably when unknown samples are semantically similar to known classes. A significant limitation is the substantial computational burden these methods impose when applied to large datasets or deployed in real-time operational environments.

2.2. Discriminative methods

Previous discriminative methods utilise classifier scores to evaluate the similarity between samples and known classes, often employing conventional machine learning techniques Bendale and Boult (2015). Scheirer et al. (2013) formalised OSR problems and utilised a support vector machine for discrimination. Hendrycks and Gimpel (2017) proposed the first deep learning-based OSR method, which identified unknown samples by negating the maximum Softmax probability. Bendale and Boult (2015) reported that the max-Softmax approach for OSR exhibits limitations in classification and proposed OpenMax, which offered improved discriminative capability by recalibrating Softmax scores and applying a threshold on the maximum probability to reject unknown samples. Zhou et al. (2021) introduced the concept of placeholder learning, which calibrates overly confident predictions by reserving classifier placeholders for unknown classes. Xu et al. (2023) employed supervised contrastive learning to enhance the robustness and efficacy of OSR tasks.

Compared with generative methods, discriminative methods are generally more reliable and computationally efficient, but they can exhibit inferior performance in certain scenarios. Existing discriminative studies predominantly employ CNNs for image feature extraction and representation, which in turn potentially limit the ability to capture the global context of images. The model proposed in this study adopts a discriminative approach.

3. Preliminary

3.1. Open set recognition

OSR is a task in which a model must accurately classify instances from known classes while identifying instances from previously unattested classes as unknown. Formally, given a dataset $\mathcal{D} = \{\mathcal{X}, \mathcal{Y}\}$ containing M samples and N classes, where each instance $x_i \in \mathcal{X} := \{x_1, \cdots, x_M\}$, corresponds to a label $y_i \in \mathcal{Y} := \{y_1, \cdots, y_N\}$, the dataset can be partitioned into a known subset $\{\mathcal{X}_{known}, \mathcal{Y}_{known}\}$, such that $\mathcal{X}_{known} \subsetneq \mathcal{X}$ and $\mathcal{Y}_{known} \subsetneq \mathcal{Y}$, and an unknown testing subset, labelled $\mathcal{X}_{unknown}$. The objective of OSR is to design a classifier $f : \mathcal{X} \to \mathcal{Y}_{known} \cup \{unknown\}$, where $\mathcal{X}$ denotes the input space, and $\mathcal{Y}_{known}$ represents the label space for known classes.

To achieve this objective, a scoring function is defined to measure the confidence that a sample x belongs to one of the known classes. Letting $s_y(x)$ denote the score for class y, a decision boundary τ is then

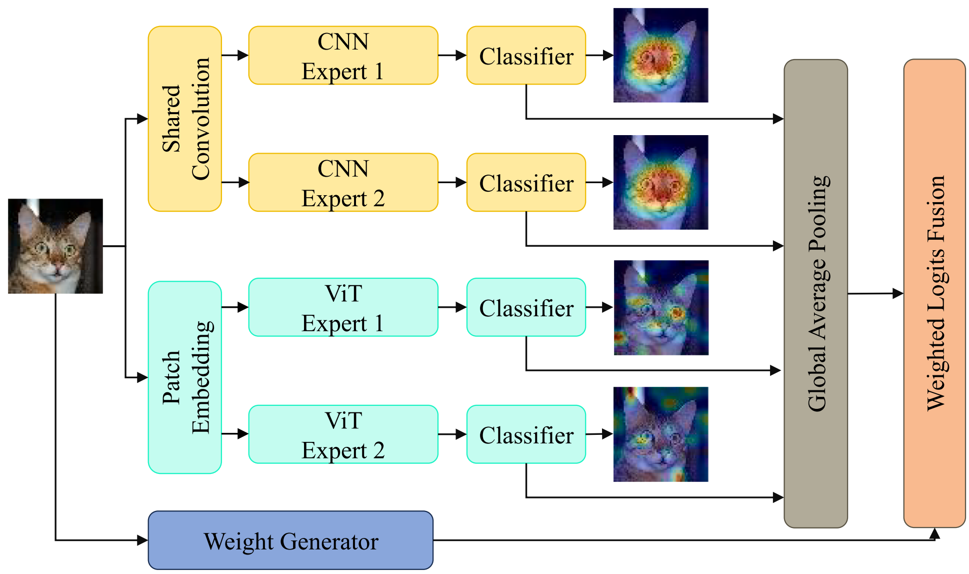

Fig. 1. Structure of the proposed model. A multi-expert system comprising both ViT and CNN experts. The CNN experts mainly focus on local features of the entity, while the ViT experts attend to global image information. A weight generator network is utilised to adaptively produce weights for integrating expert-specific predictions.

defined as follows:

$$f(x) = \begin{cases} \arg\max_{y\in\mathcal{Y}} s_y(x) & \text{if } \max_{y\in\mathcal{Y}} s_y(x) \geq \tau \\ \text{unknown} & \text{if } \max_{y\in\mathcal{Y}} s_y(x) < \tau \end{cases} \tag{1}$$

Misclassifying samples within $\mathcal{X}_{known}$ increases close-set risks $\mathcal{R}_C$, whereas accepting unknown samples from $\mathcal{X}_{unknown}$ increases open-set risks $\mathcal{R}_O$. Scheirer et al. (2013) aimed to optimise OSR tasks by minimising open-set risk. Considering the difficulty of obtaining precise information about unknown samples during training, the potential risk of the test set can be used to evaluate the performance of the classifier $f(x)$. With α representing the fraction of unknown samples, the total potential risk $\mathcal{R}_T$ can be articulated as follows:

$$\mathcal{R}_T = \alpha \cdot \mathcal{R}_O(\mathcal{X}_{unknown}, f(x)) + (1-\alpha) \cdot \mathcal{R}_C(\mathcal{X}_{known}, f(x)) \tag{2}$$

3.2. Vision transformer

ViTs have emerged as a transformative approach in computer vision, shifting the paradigm from conventional CNNs to transformer-based models. First introduced by Dosovitskiy et al. (2020), ViTs leverage the self-attention mechanism originally developed for natural language processing. Given sufficient training data, ViTs have been shown to outperform CNNs on a variety of image classification benchmarks, including ImageNet.

The partitioning of images into fixed-size patches, which are subsequently transformed into position-embedded tokens, facilitates their effective integration into the attention mechanism of the transformer. This architectural design allows ViTs to capture long-range dependencies and global contextual information more effectively compared with CNNs, which are inherently constrained by their localised receptive fields.

4. Proposed model

Following the mixture-of-experts framework, as detailed by Shazeer et al. (2017), a model architecture comprising multiple specialised experts was designed for the task. The core principle of the network architecture is that the shallow layers (responsible for early feature extraction) are shared across multiple experts, while the deeper layers (which capture more complex, task-specific features) are individualised for each expert. This architecture allows the network to leverage both shared representations and expert specialisation, which is particularly beneficial in multitask or multi-class learning scenarios. The architecture incorporates shared parameters among experts in the initial layers, transitioning to individualised parameters in the deeper layers. This design enables each expert to specialise in capturing specific semantic features.

Fig. 1 shows the architecture of the proposed model, comprising two CNN experts and two ViT experts that process images in parallel. Each training sample (x, y) is processed through both the CNN and ViT expert pathways, which in turn capture different discriminative aspects of the image.

4.1. Convolutional neural network expert

Significant advancements in image processing have been catalysed by the synergistic combination of convolutional kernels and residual neural networks. Within the domain of deep learning, architectures such as ResNet He et al. (2016) have demonstrated exceptional proficiency in extracting relevant features from images. At its core, the residual unit constitutes the fundamental component of a residual network, which can be formulated as follows:

$$x'^n = BN(Conv(F(BN(Conv(x^n))))) \tag{3}$$

$$x^{n+1} = F(x'^n + Shortcut(x^n)) \tag{4}$$

where x represents the input features, $Shortcut(x)$ corresponds to an exact replication of the original x, and $Conv(x)$ and $BN(x)$ represent convolution and batch normalization operations, respectively. $F(x)$ represents a non-linear activation function, such as ReLU or tanh.

A CNN is divided into shared shallow layers and independent deeper layers. The shared shallow layers capture general features common across all tasks or classes. They perform the initial feature extraction (e.g., edge detection and texture recognition) and are shared by all experts, which ultimately reduces the number of parameters and prompts the model to acquire more generalisable features. The independent deeper layers are specialised for each expert, thereby allowing experts to learn representations tailored to their specific task. This specialisation enables the model to capture task-specific complexities, making each expert capable of focusing on the nuances of different tasks or data distributions.

In the proposed architecture, the image is first processed through multiple shared shallow convolutional layers, each composed of several residual units. Afterwards, copies of the image are distributed to

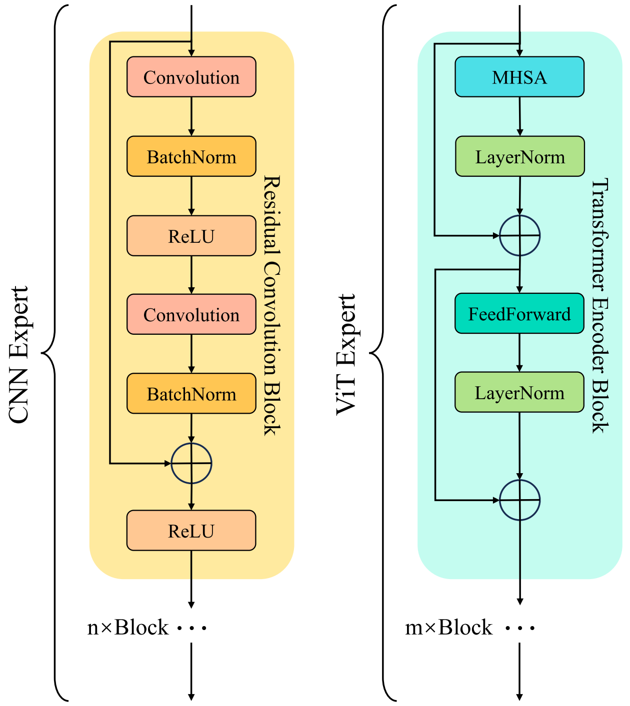

Fig. 2. Structure of the CNN expert and ViT expert. Each expert consists of multiple cascaded basic blocks.

different CNN experts, each comprising additional residual units (Fig. 2). At the output stage, each expert generates its predictions, or logits, representing its specialised decision.

4.2. Vision transformer expert

In ViTs, images are initially divided into patches and subsequently tokenised into tokens x. To encode the spatial relationships among these image patches, positional embeddings are appended to each token, thereby facilitating the learning of their 2D spatial arrangement. According to Eq. 5, x_c represents the class token, and x_p denotes the positional embeddings.

$$x = (x_1, x_2, \dots, x_m; x_c) + x_p \tag{5}$$

$$x'^n = x^n + MHSA(LayerNorm(x^n)) \tag{6}$$

$$x^{n+1} = x'^n + FF(LayerNorm(x'^n)) \tag{7}$$

After the patch embedding procedure, the tokens are processed by transformer encoders (or transformer blocks). Each encoder mainly comprises two key sub-layers, as follows: the multi-head self-attention layer and the feed-forward (FF) network (Eq. 6 and 7). Transformer models consist of cascaded transformer blocks, where x^n represents the output after the n-th encoder block (Fig. 2).

In the proposed architecture, an image is first passed through the shared patch embedding procedure and tokenised. Afterwards, its copies are distributed to different ViT experts to learn intrinsic, particularly global, features of the image. Finally, each expert outputs the logits for all learned close-set categories.

4.3. Weight generator and weighted fusion

The weight generator, as proposed in the architecture of Shazeer et al. (2017), is a simple classification network with an independent loss function for generating weights for each expert. In this work, a separate network (structurally similar to the initial shared layers and a single expert path) acts as the weight generator, producing weights W for each expert. This design provides granular control over each expert's contribution to the final result, as the weight generator considers features and correlations independent of the experts. For every input x, the trainable weight generator network processes the sample. The network only retains the values for the top k experts, setting all other values to $-\infty$. Applying the Softmax function to the top values produces a set of continuous weights for the selected experts, while all remaining experts receive a weight of zero. The weight generator network is jointly trained with the rest of the model via backpropagation.

The final logits, denoted as l, are computed by combining the logits from each expert, l_i, with their corresponding weights, w_i, generated by the weight generator, as follows:

$$l = \sum_{i=1}^{N} w_i \times l_i \tag{8}$$

The overall loss $\mathcal{L}$ consists of expert-wise cross-entropy loss and weight generator loss. Letting $\mathcal{L}_v^i$ denote the loss from the i-th ViT expert and $\mathcal{L}_c^j$ denote the loss from the j-th CNN expert, the overall loss is expressed as follows:

$$\mathcal{L} = \mathcal{L}_{wg} + \beta_1 \times \sum_{i=1} \mathcal{L}_v^i + \beta_2 \times \sum_{j=1} \mathcal{L}_c^j \tag{9}$$

where $\mathcal{L}_{wg}$ denotes the cross-entropy loss of the weight generator, and β_1, β_2 are scalable factors.

The cross-entropy loss of the i-th expert can be formulated as follows:

$$\mathcal{L}_{ce}^i = -\sum_k y_k \times \log(Softmax(l_i)^k) \tag{10}$$

where y_k represents the true label (ground truth) for the k-th class, and l_i represents the logits output of the i-th expert.

4.4. Rejecting unknowns

The open space risk $\mathcal{R}_O$ (Eq. 2) indicates that the probability of correctly rejecting an unknown sample is inversely proportional to the amount of shared information between its attributes and those of any known class.

A major challenge arises in OSR when unknown instances exhibit "overlapping features" or partial similarities with known categories. For example, an unknown sample may possess local textures or patterns similar to a known class, causing a standard classification network to produce erroneously high confidence scores (logits). To mitigate this challenge, a feature-consistency verification mechanism is implemented to neutralise the misleading impact of these partial similarities. Feature maps from the final layer of each expert are utilised to account for overlapping features.

Let M_i denote the feature map from the final layer of the i-th expert. For a CNN expert, this map corresponds to the output of the final convolutional layer (x^{n+1} in Eq. 4). For a ViT expert, this map corresponds to the output of the final FF network layer (x^{n+1} in Eq. 7). The L2 norm of the averaged feature maps across n experts can be expressed as follows:

$$S_{feature}(x) = \left\| \frac{\sum_{i=1}^{n} M_i}{n} \right\|_2 \tag{11}$$

Given that $S_{logits}(x)$ represents the maximum value of the experts' logits outputs, γ denotes the proportion of $S_{feature}(x)$ contributing to the composition of the final scoring function $S(x)$ formulated as follows:

$$S(x) = S_{logits}(x) + \gamma \cdot S_{feature}(x) \tag{12}$$

With the scoring function $S(x)$, the decision boundary τ is determined by accepting 95% of known samples (Eq. 1). Thus, τ corresponds to the maximum value of the lowest 5% of $S(x)$ scores. Samples with

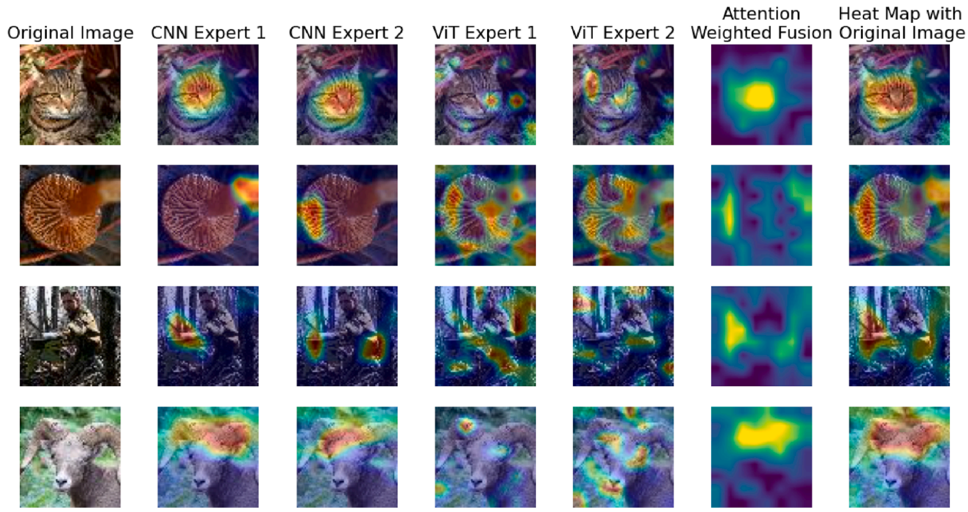

Fig. 3. Visualization of experts' attention on images with heatmap. The CVEF model consists of 2 CNN experts and 2 ViT experts. The CNN experts always have a local attention to certain textural features of images while ViT experts have attention over the whole image with a better understanding of global information and features. By combining two approaches via a weighted fusion, the model can have a better understanding of the image.

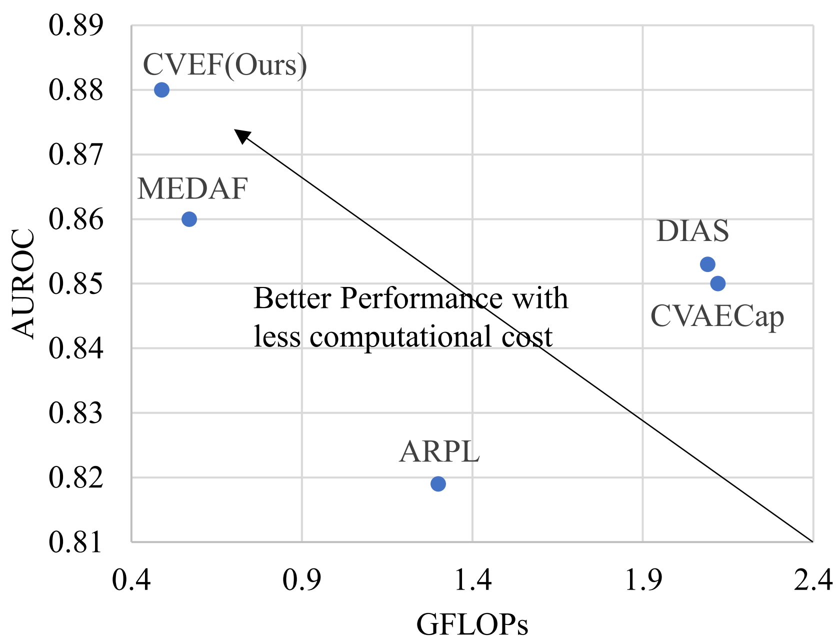

Fig. 4. The computational cost (measured in GFLOPS) and performance (AUROC) of the models on the CIFAR10 experiment are compared. Models aiming for high AUROC with low GFLOPS are preferred.

scores below the threshold τ are subsequently classified as unknown, as follows:

$$\tau = \max\{S(x) : P(S(x) \leq S(X_{known})) \leq 0.05\} \tag{13}$$

5. Experiments

5.1. Detailed implementation

In the experiments conducted in this study, ResNet18 He et al. (2016) serves as the backbone for the CNN experts, while a ViT-Base model, configured with three encoder blocks and a patch size of 8, is employed as the ViT expert architecture. For optimisation, the AdamW Loshchilov and Hutter (2018) optimiser is utilised with a learning rate of 0.001, $\beta_1 = 0.9$, $\beta_2 = 0.99$ and a weight decay of 10^{-5}. With a fixed batch size of 128, setting $\gamma = 0.01$, the model was trained on an Nvidia RTX 4090 with CUDA 12.1 for 300 epochs. The comparative analysis directly adopts the reported results for methods with identical configurations. In cases of discrepancies in settings, results reproduced from the respective official codebases were utilised. Methods without publicly available source code were represented solely by the experimental outcomes provided by their original authors.

5.2. Unknown detection

This section evaluates the model's capability to identify samples from previously unattested classes, across several image datasets. Following the evaluation protocol described in Moon et al. (2022), the experiments were conducted using four image datasets: CIFAR10 Krizhevsky (2009), CIFAR + 10, CIFAR + 50 Netzer et al. (2011), and Tiny-ImageNet Le and Yang (2015). The area under the receiver operating characteristic (AUROC) is a threshold-independent metric used to assess

Table 1
Comparative performance of methods on unknown detection tasks (AUROC). All results are the average values, and the best results are in Bold.

Method	CIFAR10	CIFAR+10	CIFAR+50	Tiny-ImageNet
Softmax Hendrycks and Gimpel (2017)	0.677	0.816	0.805	0.577
C2AE Oza and Patel (2019)	0.711	0.810	0.803	0.581
CGDL Sun et al. (2020)	0.681	0.794	0.794	0.653
RPL Chen et al. (2020)	0.784	0.885	0.881	0.711
PROSER Zhou et al. (2021)	0.801	0.898	0.881	0.684
CVAECap Guo et al. (2021)	0.835	0.888	0.889	0.715
ARPL Chen et al. (2021)	0.819	0.904	0.901	0.710
NAS-OSR Zhang et al. (2022)	0.843	0.840	0.871	-
DIAS Moon et al. (2022)	0.850	0.920	0.916	0.731
MEDAF Wang et al. (2024)	0.860	0.960	0.955	0.800
CVEF(Ours)	**0.880**	**0.966**	**0.959**	**0.802**

the model's capability in distinguishing known classes from unknown ones. To ensure a fair comparison among various methods, identical dataset splits were adopted from Chen et al. (2021), Moon et al. (2022), Neal et al. (2018), Wang et al. (2024). The datasets were configured as follows: For CIFAR10, six categories were randomly selected as known classes, while the remaining four were designated as unknowns. Similarly, TinyImageNet's structure involved designating 20 classes as known and employing the subsequent 180 classes as anomalies. For the CIFAR10 and CIFAR+50 datasets, the training regimen involved four non-animal categories from CIFAR10 as the known set. Correspondingly, a pool of 10 or 50 animal categories, respectively, was randomly sampled from CIFAR100 to comprise the unknown elements.

Table 1 compares the AUROC performance against both generative and discriminative methods. As demonstrated by the experimental outcomes, CVEF typically achieves superior performance compared to both generative and discriminative OSR approaches. Among the evaluated methods, CVEF consistently outperforms prior approaches across most datasets, achieving the highest AUROC scores. This comprehensive superiority in AUROC metrics across varied tasks and datasets underscores CVEF's robustness and efficacy in detecting unknown samples, demonstrating CVEF's effectiveness at distinguishing between known and unknown classes.

5.3. Open set recognition

OSR presents a dual challenge: accurately categorising known classes while simultaneously identifying and rejecting unknown instances. The testing methodology utilised CIFAR10 data to establish a baseline of known categories during the training phase. For evaluation purposes, experiments introduced external images sourced from both ImageNet Russakovsky et al. (2015) and LSUN Yu et al. (2016) as unknown elements, following the protocol of Yoshihashi et al. (2019). To maintain dimensional consistency with the CIFAR10 reference set, these external images underwent cropping or resizing. To maintain a consistent test sample size of exactly 10,000 images, four distinct evaluation datasets were created: the cropped and resized variants of ImageNet (IMGN-C and IMGN-R) and the corresponding modifications of LSUN (LSUN-C and LSUN-R). To facilitate an equitable comparison, the four datasets provided by Liang et al. (2018) were utilised. Model discrimination ability was assessed using macro-averaged F1-scores calculated across 11 classes: 10 used for training and one unseen. Table 2 illustrates that the proposed method achieves a substantial performance improvement over current generative and discriminative methods.

Table 2
Comparative Macro-F1 performance on CIFAR10 with ImageNet(Crop, Resize) and LSUN(Crop, Resize).

Method	IMGN-C	IMGN-R	LSUN-C	LSUN-R
Softmax Hendrycks and Gimpel (2017)	0.639	0.653	0.642	0.647
CROSR Yoshihashi et al. (2019)	0.721	0.735	0.720	0.749
C2AE Oza and Patel (2019)	0.837	0.826	0.783	0.801
GFROSR Perera et al. (2020)	0.757	0.792	0.751	0.805
CGDL Sun et al. (2020)	0.840	0.832	0.806	0.812
PROSER Zhou et al. (2021)	0.849	0.824	0.867	0.856
ConOSR Xu et al. (2023)	0.891	0.843	0.912	0.881
MEDAF Wang et al. (2024)	0.915	0.900	0.922	0.926
CVEF(Ours)	**0.930**	**0.925**	**0.930**	**0.941**

5.4. Out of distribution detection

In line with the experimental setup outlined by Chen et al. (2020), experiments utilised two pairs of out-of-distribution (OOD) detection benchmarks Hendrycks and Gimpel (2017) involving SVHN, CIFAR10, and CIFAR100. Specifically, the model is trained on CIFAR10. CIFAR100 is then employed as the similar OOD dataset, while SVHN functions as the OOD dataset with a significant difference from the trained data to assess the model's performance against varying degrees of distributional shift.

Detection Accuracy (DTACC) is a metric that quantifies the maximum classification accuracy between known and unknown classes across all possible threshold values. The calculation of this accuracy operates under the assumption of equiproportional representation of positive and negative samples within the test set. **Area Under the Precision-Recall Curve (AUPR)** serves as a performance metric by plotting precision, defined as $TP/(TP + FP)$, against recall, defined as $TP/(TP + FN)$, across varying threshold values. The terms TP, FP, and FN refer to true positives, false positives, and false negatives, respectively. AUPR is additionally segmented into AUIN and AUOUT. For the computation of AUIN, in-distribution samples are treated as positive, while for AUOUT, out-of-distribution samples are regarded as positive.

Table 3 demonstrates that the proposed method achieves results highly comparable to the current state-of-the-art for the CIFAR10 to CIFAR100 configuration. Similarly, for the CIFAR10/SVHN setup, CVEF achieves superior performance. These results indicate the effectiveness of CVEF in distinguishing in-distribution and out-of-distribution samples, showcasing its robustness and reliability across diverse datasets.

5.5. Ablation study

As shown in Table 4, the AUROC on the TinyImageNet task with is evaluated with different methods and composition structures. The results indicate that the weight generator is instrumental in adaptively fusing expert outputs. Individual architectures, such as a "pure CNN" or "pure ViT," exhibit inferior performance compared with the proposed approach, achieving a synergistic effect where the combined performance exceeds that of the individual components. CNN and ViT experts provide complementary strengths, thereby enhancing overall performance.

Table 3
Out-of-distribution performance on CIFAR10 with similar(CIFAR100) and different(SVHN) OOD datasets.

Method	Known:CIFAR10/Unknown:CIFAR100				Known:CIFAR10/Unknown:SVHN			
	DTACC	AUROC	AUIN	AUOUT	DTACC	AUROC	AUIN	AUOUT
Softmax Hendrycks and Gimpel (2017)	0.789	0.863	0.884	0.825	0.864	0.906	0.883	0.936
RPL Chen et al. (2020)	0.806	0.871	0.888	0.838	0.871	0.920	0.896	0.951
CSI Tack et al. (2020)	0.844	0.916	0.925	0.900	0.928	0.979	0.962	0.990
GCPL Yang et al. (2022)	0.802	0.864	0.866	0.841	0.861	0.913	0.866	0.948
ARPL Chen et al. (2021)	0.834	0.903	0.911	0.884	0.916	0.966	0.948	0.980
OpenGAN Kong and Ramanan (2021)	0.842	0.897	0.877	0.896	0.921	0.959	0.934	0.971
MEDAF Wang et al. (2024)	0.854	0.925	0.932	0.911	0.953	0.991	0.980	0.996
CVEF(Ours)	0.853	0.921	0.927	0.907	0.960	0.992	0.985	0.997

Table 4
Performance with different setups of CVEF, with AUROC on TinyImageNet. The default CVEF model of this work consists of 2 CNN experts and 2 ViT experts.

Method	AUROC
CVEF (2 ViT and 2 CNN experts)	0.802
CVEF without Weight Generator	0.796
CVEF with 4 CNN experts	0.763
CVEF with 1 ViT and 1 CNN expert	0.710
CVEF with 4 ViT experts	0.706
Plain CNN	0.691

Table 5
AUROC performance with different γ on CIFAR10.

γ	AUROC
0	0.875
0.001	0.877
0.01	0.880
0.1	0.853
1	0.663

Table 6
Close-set classification performance comparison with TinyImageNet.

Method	CIFAR10	CIFAR100	Tiny-ImageNet
Plain CNN	0.940	0.716	0.637
ARPL Chen et al. (2021)	0.941	0.721	0.657
ConOSR Xu et al. (2023)	0.946	0.730	0.661
CVEF(Ours)	0.957	0.767	0.710

Table 5 shows the effectiveness of combining feature scores with logit scores for rejecting unknowns. By adjusting the ratio γ of feature scores in the overall score $S(x)$, the model can achieve better performance in distinguishing between known and unknown samples. An excessively high γ value may lead to overemphasis on feature scores, potentially neglecting important information from logit scores, resulting in a decline in performance.

5.6. Further analysis

Close-set Accuracy. To assess the model's close-set capabilities, training was conducted on CIFAR10, CIFAR100, and the initial 100 classes of TinyImageNet. As illustrated in Table 6, the proposed model outperforms baseline approaches, thereby underscoring its robust potential in both close-set and open-set contexts.

Visualisation. Fig. 3 presents a visualisation of different experts' attention to certain images using the CAM Zhou et al. (2016) technique. The visualisation reveals that CNN experts predominantly focus on localised textural features, whereas ViT experts exhibit global attention over the entire image, which may potentially result in spurious predictions. Parallel experts of the same kind may have similar attention maps or comprehension patterns, but they are still not exactly the same. Depending on the characteristics of the instances, the weight generator gives different experts different weights and fuses them adaptively.

By combining two approaches in a multi-expert system with an adaptive weighted fusion, the model leverages the strengths of both approaches. The proposed model exhibits both local attention to salient textural features of an image and global contextual understanding, thus achieving better performance in classification and discrimination.

Computational Cost. To evaluate the efficiency of the model, Fig. 4 compares the computational cost and performance on the CIFAR10 experiment. It shows that the CVEF method achieves competitive performance with a low computational cost of 0.489 GFLOPS. As mentioned in 2.1, generative methods usually have higher computational costs, as also shown in the figure, which indicates that they are not suitable for large-scale or real-time scenarios.

6. Conclusion

This paper presents the CVEF model, a multi-expert system developed for OSR. The proposed model integrates the complementary strengths of CNNs and ViTs to effectively address the challenges of identifying both known and unknown categories within complex and dynamic datasets. By combining the global contextual understanding of ViTs with the localised feature extraction capabilities of CNNs, CVEF produces a balanced and robust feature representation. The weight generator within CVEF dynamically integrates predictions from multiple experts, enhancing adaptability across diverse datasets and experimental settings. This modular architecture supports scalability to more complex tasks and larger datasets.

Experiments on multiple standard benchmark datasets demonstrate that the proposed method outperforms existing generative and discriminative approaches. Despite the integration of CNNs and ViTs, the proposed framework maintains computational efficiency, offering a practical solution for real-world applications requiring dynamic handling of known and unknown categories. High-dimensional data and large-scale image datasets continue to pose significant challenges for OSR in real-world applications. Future work will explore OSR problems involving hierarchical class structures. Moreover, the imbalance in convergence speeds between ViTs and CNNs presents a challenge in this work, as ViTs require more training epochs, causing CNNs to dominate early-stage learning and thereby hindering balanced feature integration. Future work may explore techniques such as dynamic loss weighting or pretraining ViTs to harmonise optimisation and fully leverage the strengths of both architectures.

CRediT authorship contribution statement

Kai Ding: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization; **Yu Mao:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Investigation, Funding acquisition; **Hui Chen:** Writing – review & editing, Validation, Supervision; **Yaojin Lin:** Writing – review & editing, Supervision, Software, Resources, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Bendale, A., & Boult, T. (2015). Towards open world recognition. In *2015 IEEE Conference on computer vision and pattern recognition (CVPR)* (pp. 1893–1902). https://doi.org/10.1109/CVPR.2015.7298799

Chen, G., Peng, P., Wang, X., & Tian, Y. (2021). Adversarial reciprocal points learning for open set recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, (pp. 1–1). https://doi.org/10.1109/TPAMI.2021.3106743

Chen, G., Qiao, L., Shi, Y., Peng, P., Li, J., Huang, T., Pu, S., & Tian, Y. (2020). Learning open set network with discriminative reciprocal points. In A. Vedaldi, H. Bischof, T. Brox, & J.-M. Frahm (Eds.), *Computer vision – ECCV 2020* Lecture Notes in Computer Science (pp. 507–522). Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-58580-8_30

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2020). An image is worth 16x16 words: transformers for image recognition at scale. In *International conference on learning representations*.

Ge, Z., Demyanov, S., & Garnavi, R. (2017). Generative openmax for multi-Class open set classification. In *British machine vision conference 2017, BMVC 2017, london, UK, september 4–7, 2017*. BMVA Press.

Geng, C., Huang, S.-J., & Chen, S. (2021). Recent advances in open set recognition: a survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *43*(10), 3614–3631. https://doi.org/10.1109/TPAMI.2020.2981604

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. In *Advances in neural information processing systems*. Curran Associates, Inc. (*vol. 27*).

Guo, Y., Camporese, G., Yang, W., Sperduti, A., & Ballan, L. (2021). Conditional variational capsule network for open set recognition. In *2021 IEEE/CVF International conference on computer vision (ICCV)* (pp. 103–111). https://doi.org/10.1109/ICCV48922.2021.00017

He, K., Zhang, X., Ren, S., & Sun, J. (2015). Delving deep into rectifiers: surpassing human-Level performance on imagenet classification. In *2015 IEEE International conference on computer vision (ICCV)* (pp. 1026–1034). https://doi.org/10.1109/ICCV.2015.123

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770–778).

Hendrycks, D., & Gimpel, K. (2017). A baseline for detecting misclassified and out-of-Distribution examples in neural networks. In *International conference on learning representations*.

Kingma, D. P., & Welling, M. (2014). Auto-encoding variational bayes. In Y. Bengio, & Y. LeCun (Eds.), *2Nd international conference on learning representations, ICLR 2014, banff, AB, canada, april 14–16, 2014, conference track proceedings*.

Kong, S., & Ramanan, D. (2021). OpenGAN: open-Set recognition via open data generation. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 813–822).

Krizhevsky, A. (2009). Learning multiple layers of features from tiny images.

Le, Y., & Yang, X. (2015). Tiny imagenet visual recognition challenge. *CS 231N*, *7*(7), 3.

Liang, S., Li, Y., & Srikant, R. (2018). Enhancing the reliability of out-of-distribution image detection in neural networks. In *International conference on learning representations*.

Loshchilov, I., & Hutter, F. (2018). Decoupled weight decay regularization. In *International conference on learning representations*.

Moon, W., Park, J., Seong, H. S., Cho, C.-H., & Heo, J.-P. (2022). Difficulty-aware simulator for open set recognition. In S. Avidan, G. Brostow, M. Cissé, G. M. Farinella, & T. Hassner (Eds.), *Computer vision – ECCV 2022* (pp. 365–381). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-19806-9_21

Neal, L., Olson, M., Fern, X., Wong, W.-K., & Li, F. (2018). Open set learning with counterfactual images. In V. Ferrari, M. Hebert, C. Sminchisescu, & Y. Weiss (Eds.), *Computer vision – ECCV 2018* (pp. 620–635). Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-01231-1_38

Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., & Ng, A. Y. (2011). Reading digits in natural images with unsupervised feature learning. In *NIPS Workshop on deep learning and unsupervised feature learning* (p. 4). (*vol. 2011*).

Oza, P., & Patel, V. M. (2019). C2AE: Class conditioned auto-Encoder for open-Set recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 2307–2316).

Perera, P., Morariu, V. I., Jain, R., Manjunatha, V., Wigington, C., Ordonez, V., & Patel, V. M. (2020). Generative-discriminative feature representations for open-Set recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 11814–11823).

Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., Huang, Z., Karpathy, A., Khosla, A., Bernstein, M., Berg, A. C., & Fei-Fei, L. (2015). Imagenet large scale visual recognition challenge. *International Journal of Computer Vision*, *115*(3), 211–252. https://doi.org/10.1007/s11263-015-0816-y

Scheirer, W. J., de Rezende Rocha, A., Sapkota, A., & Boult, T. E. (2013). Toward open set recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *35*(7), 1757–1772. https://doi.org/10.1109/TPAMI.2012.256

Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., & Dean, J. (2017). Outrageously large neural networks: the sparsely-Gated mixture-of-Experts layer. In *International conference on learning representations*.

Sun, X., Yang, Z., Zhang, C., Ling, K.-V., & Peng, G. (2020). Conditional gaussian distribution learning for open set recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 13480–13489).

Tack, J., Mo, S., Jeong, J., & Shin, J. (2020). CSI: Novelty detection via contrastive learning on distributionally shifted instances. In *Advances in neural information processing systems* (pp. 11839–11852). Curran Associates, Inc. (*vol. 33*).

Vaze, S., Han, K., Vedaldi, A., & Zisserman, A. (2021). Open-set recognition: a good closed-Set classifier is all you need. *ArXiv*, .

Wang, Y., Mu, J., Zhu, P., & Hu, Q. (2024). Exploring diverse representations for open set recognition. *Proceedings of the AAAI Conference on Artificial Intelligence*, *38*(6), 5731–5739. https://doi.org/10.1609/aaai.v38i6.28385

Xu, B., Shen, F., & Zhao, J. (2023). Contrastive open set recognition. *Proceedings of the AAAI Conference on Artificial Intelligence*, *37*(9), 10546–10556. https://doi.org/10.1609/aaai.v37i9.26253

Yang, H.-M., Zhang, X.-Y., Yin, F., Yang, Q., & Liu, C.-L. (2022). Convolutional prototype network for open set recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *44*(5), 2358–2370. https://doi.org/10.1109/TPAMI.2020.3045079

Yoshihashi, R., Shao, W., Kawakami, R., You, S., Iida, M., & Naemura, T. (2019). Classification-reconstruction learning for open-Set recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 4016–4025).

Yu, F., Seff, A., Zhang, Y., Song, S., Funkhouser, T., & Xiao, J. (2016). LSUN: Construction of a Large-scale Image Dataset using Deep Learning with Humans in the Loop. https://doi.org/10.48550/arXiv.1506.03365

Zhang, X., Cheng, X., Zhang, D., Bonnington, P., & Ge, Z. (2022). Learning network architecture for open-Set recognition. *Proceedings of the AAAI Conference on Artificial Intelligence*, *36*(3), 3362–3370. https://doi.org/10.1609/aaai.v36i3.20246

Zhao, X., Wang, L., Zhang, Y., Han, X., Deveci, M., & Parmar, M. (2024). A review of convolutional neural networks in computer vision. *Artificial Intelligence Review*, *57*(4), 99. https://doi.org/10.1007/s10462-024-10721-6

Zhou, B., Khosla, A., Lapedriza, A., Oliva, A., & Torralba, A. (2016). Learning deep features for discriminative localization. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 2921–2929).

Zhou, D.-W., Ye, H.-J., & Zhan, D.-C. (2021). Learning placeholders for open-Set recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 4401–4410).